



UNIVERSITY OF
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Superconductivity

Lecture 11 – “Invariance & Quantisation”

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Previously, Strong Fields

The full GL equation,

$$\frac{1}{2m^*} (-i\hbar\nabla + e^*\mathbf{A})^2 \psi + a\psi + b\psi|\psi|^2 = 0$$

could be linearised for $\psi \simeq 0$.

$$\mu_0 H_{c2} = \frac{\Phi_0}{2\pi\xi(T)^2}$$

$$\kappa = \lambda/\xi$$

- $\kappa < 1/\sqrt{2}$ - type I
- $\kappa > 1/\sqrt{2}$ - type II

Experimental fluoxid quantisation

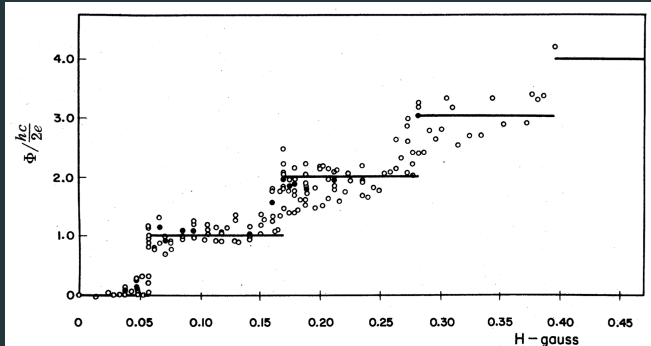


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Complex order parameter

Phase term $e^{i\theta(\mathbf{r})}$

Complex order parameter,

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\theta(\mathbf{r})}.$$

The total free energy,

$$F[\psi, \mathbf{A}] = \int_V d^3\mathbf{r} \left[a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi(\mathbf{r})|^2 + \frac{B^2}{2\mu_0} \right],$$

where the last term was added to account for the field energy.

Which of the terms above relevant to $\theta(\mathbf{r})$?

$$\frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi(\mathbf{r})|^2$$

Free energy

Use

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\theta(\mathbf{r})}.$$

to evaluate

$$\begin{aligned}\frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi(\mathbf{r})|^2 &= \frac{1}{2m^*} |(-i\hbar\nabla|\psi| + \hbar|\psi|\nabla\theta + e^*\mathbf{A}|\psi|)e^{i\theta}|^2 \\ &= \dots \\ &= \frac{\hbar^2}{2m^*} (\nabla|\psi|)^2 + \frac{1}{2m^*} |\psi|^2 (\hbar\nabla\theta + e^*\mathbf{A})^2\end{aligned}$$

On the left,

- 1st term: amplitude variation
- 2nd term: phase variation (kinetic energy; supercurrent)

Super-current

Minimise F (1/2)

$$F[\psi, \mathbf{A}] = \int_V d^3\mathbf{r} \left[a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi(\mathbf{r})|^2 + \frac{B^2}{2\mu_0} \right],$$

Find the condition where $\delta F = 0$ for any $\delta\mathbf{A}$ (F minimum):

1. Evaluate $F + \delta F$ by replacing $\mathbf{A}$ with $(\mathbf{A} + \delta\mathbf{A})$ on the right;
2. Expand and keep only the linear term in $\delta\mathbf{A}$;
3. Express in the form of $\delta F = (\dots)\delta\mathbf{A}$;
4. $(\dots) = 0$ is the solution.

Minimise F (2/2)

One can find

$$\delta F = \int_V d^3\mathbf{r} \left[\mathbf{\blacksquare} \cdot \delta \mathbf{A} + \frac{1}{\mu_0} (\nabla \times \mathbf{A}) \cdot (\nabla \times \delta \mathbf{A}) \right]$$

where

$$\mathbf{\blacksquare} = \frac{-i\hbar e^{*2}}{2m^*} (\psi^* \nabla \psi - \psi \nabla \psi^*) + \frac{e^{*2}}{m^*} |\psi|^2 \mathbf{A}$$

This can be further rewritten as

$$\delta F = \int_V d^3\mathbf{r} \left[\mathbf{\blacksquare} + \frac{1}{\mu_0} (\nabla \times \mathbf{B}) \right] \cdot \delta \mathbf{A}.$$

So, [...] should be zero, but ... $\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$.

Thus $\mathbf{j}_s = -\mathbf{\blacksquare}$ (also the London equation).

Express supercurrent (1/2)

Use

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\theta(\mathbf{r})},$$

to expand

$$\mathbf{j}_s = \frac{i\hbar e^{*2}}{2m^*} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{e^{*2}}{m^*} |\psi|^2 \mathbf{A}.$$

Using

$$\nabla \psi = (\nabla |\psi| + i|\psi| \nabla \theta) e^{i\theta},$$

$$\psi^* \nabla \psi - \psi \nabla \psi^* = 2i|\psi|^2 \nabla \theta.$$

one can show eventually,

$$\mathbf{j}_s = -\frac{e^*}{m^*} |\psi|^2 (\hbar \nabla \theta + e^* \mathbf{A}).$$

Similar to the kinetic energy part in F we saw earlier!

Express supercurrent (2/2)

Compare

$$\mathbf{j}_s = -\frac{e^*}{m^*} |\psi|^2 (\hbar \nabla \theta + e^* \mathbf{A})$$

with

$$\mathbf{j}_s = -n^* e^* \mathbf{v}_s$$

to obtain

$$m^* \mathbf{v}_s = (\hbar \nabla \theta + e^* \mathbf{A})$$

Fluxoid quantisation

Fluxoid quantisation (1/2)

$$m^* \mathbf{v}_s = (\hbar \nabla \theta + e^* \mathbf{A})$$

Suppose a superconductor ring,

$$\oint_C m^* \mathbf{v} \cdot d\mathbf{r} = \hbar \oint_C \nabla \theta \cdot d\mathbf{r} + e^* \oint_C \mathbf{A} \cdot d\mathbf{r}$$

We'd expect

$$\oint_C \nabla \theta \cdot d\mathbf{r} = 2\pi n$$

so that

$$\hbar \oint_C \nabla \theta \cdot d\mathbf{r} = \hbar 2\pi n = nh$$

Using Stoke's theorem,

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \int_S \mathbf{B} \cdot d\mathbf{S} = \Phi$$

Fluxoid quantisation (2/2)

$$\begin{aligned}\oint_C m^* \mathbf{v} \cdot d\mathbf{r} &= \hbar \oint_C \nabla \theta \cdot d\mathbf{r} + e^* \oint_C \mathbf{A} \cdot d\mathbf{r} \\ &= nh + e^* \Phi\end{aligned}$$

Finally, it can be written as

$$\frac{1}{e^*} \oint_C \frac{m^*}{n^* e^{*2}} \mathbf{j}_s \cdot d\mathbf{r} + \Phi = n \left(\frac{h}{e^*} \right)$$

where $h/e^* \equiv \Phi_0$.

The left part is called 'fluxoid'. The fluxoid is 'quantised'.

Experimental fluoxid quantisation

